A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 6: Security, Privacy and Data Integrity

Subtopic 6.1: Data Security

Concept 6.1.4: Data Protection Measures (Encryption, Access Rights)

Total Questions: 8

Mark Schemes begin on Page 6

Question 1 | Source: 9618_s25_qp_11 (Q7.a)

7 A banker stores personal data on their work computer.

(a) The banker needs to transfer confidential data across the internet.

Identify and describe **one** method of restricting the risks posed by an unauthorised person intercepting the data whilst it is being transferred across the internet.

Method ....

Description ....

....

....

....

.... [3]

Question 2 | Source: 9618_s25_qp_13 (Q3.b)

3 A programmer is writing a program in a high-level language.

(b) The programmer needs to keep the program files secure on their computer and during electronic data transmission over the internet. The files are protected by a password.

Complete the table by identifying **one** other method of keeping the files secure during electronic data transmission **and one** other method of keeping the files secure on the computer.

State how each method protects the data.

The methods must be different.

	Method	**How the method protects the data**
during transmission	....	....
on the computer	....	....

[4]

Question 3 | Source: 9618_w24_qp_12 (Q4.c)

4 A program is written in a high-level language by a team of three programmers using an Integrated Development Environment (IDE).

(c) The file containing the final program code will be sent by email for beta testing.

Identify **one** security method that can be used to protect the program code from unauthorised access during email transfer.

Explain how your chosen method protects the program code.

Security method ....

....

Explanation ....

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....
[3]

Question 4 | Source: 9618_s24_qp_12 (Q3.a.ii)

3 An assessment board scans exam papers and stores the digitised papers on a server. Exam markers download the digitised papers to mark. The exam markers then upload the mark for each paper.

(a) The assessment board needs to make sure the data stored on the server is secure.

(ii) Identify **one** security measure that helps to protect the data when it is being transmitted to its destination. Describe how the security measure works.

Security measure ....

Description ....

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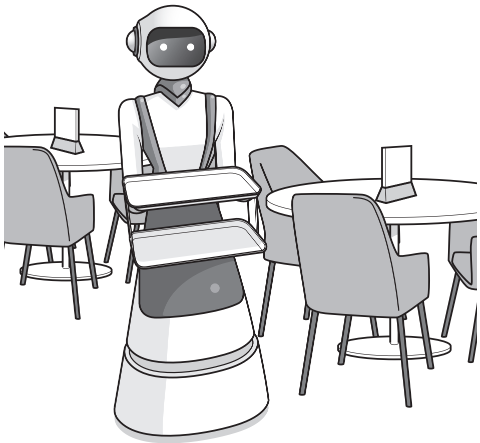
....

....
[3]

Question 5 | Source: 9618_s24_qp_13 (Q7.f.ii)

7 Robots are used to serve food and drink to customers at a restaurant.



(f) The data from the robots is transmitted to a central computer using a wireless connection.

(ii) Explain how encryption can protect the security of data during transmission.

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.... [2]

Question 6 | Source: 9618_w23_qp_12 (Q2.b)

(b) Authentication is one method a Database Management System (DBMS) can use to improve the security of a database.

Describe **other** methods that a DBMS can use to improve the security of a database.

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.... [4]

Question 7 | Source: 9618_w21_qp_11 (Q5.b)

5 Javier owns many shops that sell cars. He employs several managers who are each in charge of one or more shops. He uses the relational database `CARS` to store the data about his business.

Part of the database is shown:

```
SHOP(ShopID, ManagerID, Address, Town, TelephoneNumber)

MANAGER(ManagerID, FirstName, LastName, DateOfBirth, Wage)

CAR(RegistrationNumber, Make, Model, NumberOfMiles, ShopID)
```

(b) Describe the ways in which access rights can be used to protect the data in Javier's database from unauthorised access.

....

....

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....

....

.... [3]

Question 8 | Source: 9618_s21_qp_12 (Q8.b)

8 A company has several security measures in place to prevent unauthorised access to the data on its computers.

(b) Each employee has a username and password to allow them to log onto a computer. An employee's access rights to the data on the computers is set to either read-only, or read and write.

Identify **one** other software-based measure that could be used to restrict the access to the data on the computers.

....

.... [1]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_s25_ms_11 (Q7.a)

7(a)	**1 mark** for a correct method: **1 mark** each to **max 2** for a corresponding description Method: Encryption Description: • Data is encoded/scrambled using a key to create cipher text • if intercepted it cannot be understood • without being decrypted using a key	**3**

Mark Scheme for Question 2 | Source: 9618_s25_ms_13 (Q3.b)

3(b)	**1 mark** for method and **1 mark** for corresponding description During transmission e.g. • Encryption // by example such as VPN • Jumble / encode data so it cannot be decrypted/understood without the key On computer e.g. • Firewall / proxy • Filter incoming transmissions and stop any that could be attempting unauthorised access • Anti-malware • Find and delete or quarantine any malware that could delete the data / files • Encryption • Jumble / encode data so it cannot be decrypted / understood without the key • Physical method // by example • For example, the computer storing the data cannot be accessed without the key to the room	**4**

Mark Scheme for Question 3 | Source: 9618_w24_ms_12 (Q4.c)

4(c)	**1 mark** for the security method. **2 marks** for explanation Security method: Encryption Explanation • File contents are converted to cipher text • If intercepted the data cannot be understood without the decryption key	**3**

Mark Scheme for Question 4 | Source: 9618_s24_ms_12 (Q3.a.ii)

3(a)(ii)	**1 mark** for security measure **1 mark** each to **max 2** for description of the chosen measure: • Encryption • Encodes/scrambles data • ... so if it is intercepted it cannot be **understood** • Algorithm/key is required to decode the data	**3**

Mark Scheme for Question 5 | Source: 9618_s24_ms_13 (Q7.f.ii)

7(f)(ii)	**1 mark** each to **max 2**: • Encodes/scrambles data • ... so if it is intercepted it cannot be **understood** • Algorithm/**key** is required to decode the data	**2**

Mark Scheme for Question 6 | Source: 9618_w23_ms_12 (Q2.b)

2(b)	**1 mark** for each bullet point (**max 4**) **Max 2** if no descriptions • Backup / recovery procedures • … automatically takes copies of the database and store off site on a regular basis / weekly, etc. • ... so that the data can be recovered if lost • Use of access rights • … some users are given different access permissions to different tables • ... read/write, read only, full access, etc. • Views • … different users are able to see different parts of the database • ... only see what users need to see // by example • Record and table locking • … prevents simultaneous access to data • ... so updates are not lost // data is not overwritten • Encryption • ... the data is turned into ciphertext • ... so it cannot be understood **without a decryption key**	**4**

Mark Scheme for Question 7 | Source: 9618_w21_ms_11 (Q5.b)

5(b)	**1 mark** per bullet point • Access rights give managers / himself access to different elements • … by having different accounts / logins • … which have different access rights e.g. read only // no access / read / write • Specific <u>views</u> can be assigned to himself and to the managers • … e.g. managers can only see the data for their own shop(s)	**3**

Mark Scheme for Question 8 | Source: 9618_s21_ms_12 (Q8.b)

8(b)	**1 mark** for a correct answer • Two factor authentication • Biometric passwords • Key Card Access • Firewall	**1**